Demand Plan App — End-to-End Guide

Date: 2026-03-19 **Accounts Used:** Kroger (`0016100001IwIDPAA3`) and 84.51 (`0016100000afNAcAAM`)
Environment: http://localhost:5173 (Vite dev server) + http://localhost:8000 (FastAPI backend)

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Overview

The **Demand Plan App** is a Databricks Consumption Planning & Forecasting tool built for Field Engineering. It lets you:

- Load historical consumption data for one or more Salesforce accounts
- Visualize T12M (trailing 12-month) usage by domain and SKU
- Build 3-year consumption projections across 3 scenarios (S1/S2/S3)
- Model incremental use cases on top of baseline growth
- Export plans to Excel or upload directly to Google Drive

Prerequisites

Requirement	Details
Python 3.11+	Backend (FastAPI)
Node.js 18+	Frontend (Vite + React)
Databricks CLI	Authenticated to logfood workspace
Google OAuth	Authorized via Setup tab for Drive uploads
SFDC account IDs	Required to load account data

Deployment & Startup

1. Clone the Repository

```
git clone <repo-url>
cd demand-plan-app
```

2. Backend Setup

```
cd server
pip install -r requirements.txt
```

Ensure your Databricks CLI is authenticated to the **logfood** workspace:

```
databricks auth profiles | grep logfood
# Should show: logfood    YES
```

Start the FastAPI backend:

```
uvicorn main:app --reload --port 8000
```

The backend exposes REST endpoints at `http://localhost:8000`:

- `GET /api/setup/status` — Databricks + Google auth status
- `GET /api/accounts-search?q=<name>` — SFDC account search
- `POST /api/setup/load-account` — load historical data for an account
- `POST /api/export/upload-to-drive` — build Excel and upload to Google Drive

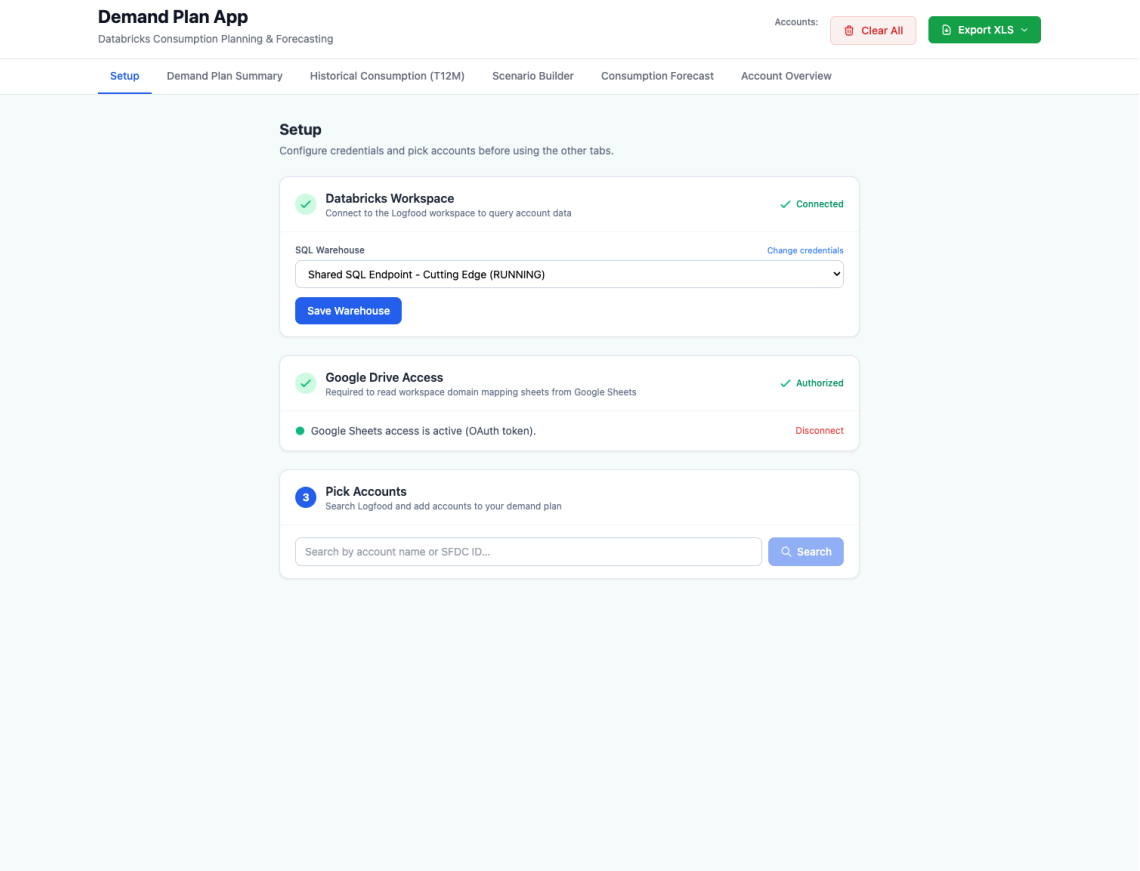
3. Frontend Setup

```
cd frontend
npm install
npm run dev
```

The React app starts at `http://localhost:5173`.

4. Open the App

Navigate to **http://localhost:5173** in Chrome. The app opens on the **Setup** tab by default.

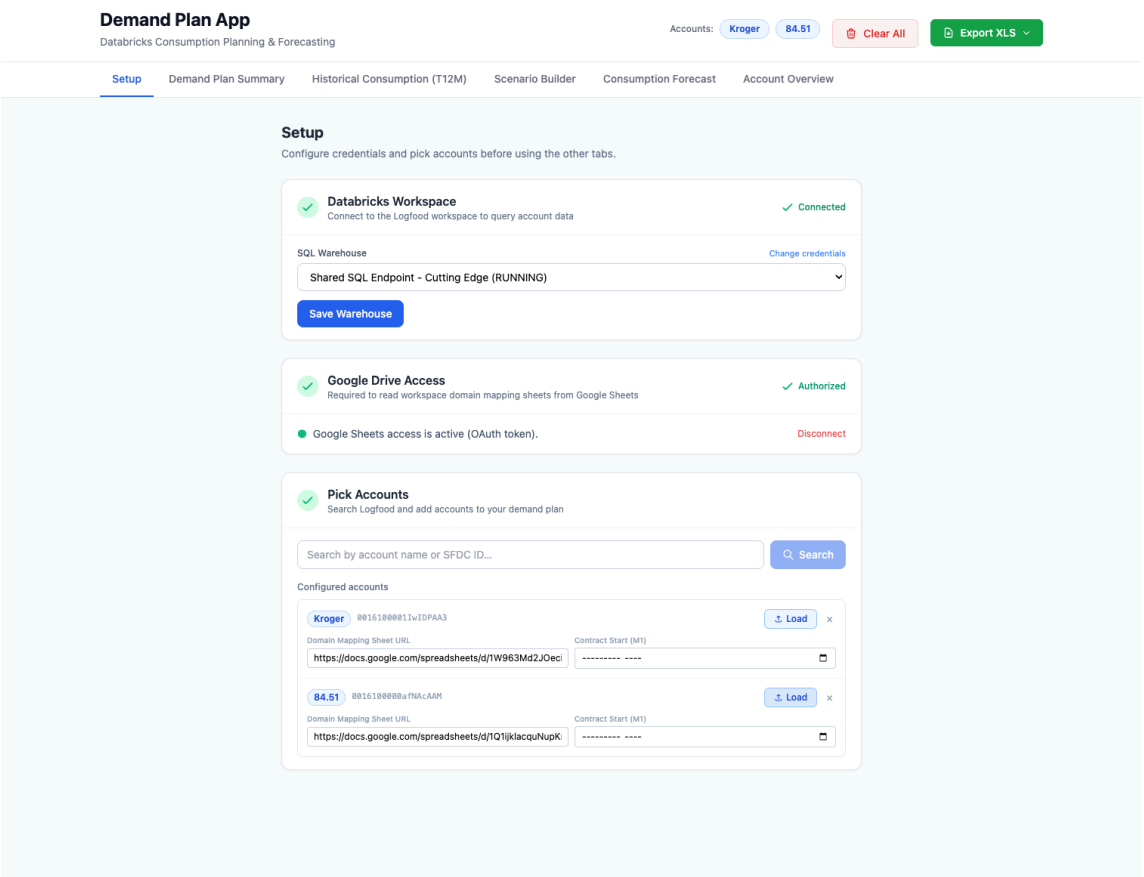


App Walkthrough

1. Setup Tab

The Setup tab is the starting point. It shows:

- **Databricks Connection** — green badge when the backend can reach the logfood SQL warehouse
- **Google OAuth** — authorization status for Google Drive uploads
- **Account Search** — search for Salesforce accounts by name and add them to your plan
- **Loaded Accounts** — list of configured accounts with their contract start dates and domain mapping URLs



Adding Accounts

1. Type an account name in the search box and click **Search**
2. Check the account checkbox in the results panel
3. Click **Add 1 account →** — the results panel auto-closes
4. Optionally set the **Domain Mapping URL** (Google Sheets link for SKU→domain mapping)
5. Click **Load** (per-account button) to fetch historical consumption from Databricks

Demand Plan App

Databricks Consumption Planning & Forecasting

Accounts: Clear All Export XLS

Setup | Demand Plan Summary | Historical Consumption (T12M) | Scenario Builder | Consumption Forecast | Account Overview

Setup

Configure credentials and pick accounts before using the other tabs.

Databricks Workspace

Connect to the Logfood workspace to query account data

Connected

SQL Warehouse — Change credentials

Shared SQL Endpoint - Cutting Edge (RUNNING)

Save Warehouse

Google Drive Access

Required to read workspace domain mapping sheets from Google Sheets

Authorized

Google Sheets access is active (OAuth token). — Disconnect

3 Pick Accounts

Search Logfood and add accounts to your demand plan

Kroger — Search

2 results

- Kroger
  0016100001IwIDPAA3
- The Kroger Co.
  001Vp00000L7nRJIAZ

Demand Plan App

Databricks Consumption Planning & Forecasting

Accounts: Kroger | Clear All | Export XLS

Setup | Demand Plan Summary | Historical Consumption (T12M) | Scenario Builder | Consumption Forecast | Account Overview

Setup

Configure credentials and pick accounts before using the other tabs.

Databricks Workspace

Connect to the Logfood workspace to query account data

✓ Connected

SQL Warehouse — Change credentials

Shared SQL Endpoint - Cutting Edge (RUNNING)

Save Warehouse

Google Drive Access

Required to read workspace domain mapping sheets from Google Sheets

✓ Authorized

● Google Sheets access is active (OAuth token). — Disconnect

Pick Accounts

Search Logfood and add accounts to your demand plan

Kroger — Search

Configured accounts

Kroger 0016100001IwIDPAA3 — Load ×

Domain Mapping Sheet URL: https://docs.google.com/spreadsheets/d/1W963Md2JOeci

Contract Start (M1): --------- ----

Demand Plan App

Databricks Consumption Planning & Forecasting

Accounts: Kroger | Clear All | Export XLS

Setup | Demand Plan Summary | Historical Consumption (T12M) | Scenario Builder | Consumption Forecast | Account Overview

Setup

Configure credentials and pick accounts before using the other tabs.

Databricks Workspace

Connect to the Logfood workspace to query account data

Connected

SQL Warehouse — Change credentials

Shared SQL Endpoint - Cutting Edge (RUNNING)

Save Warehouse

Google Drive Access

Required to read workspace domain mapping sheets from Google Sheets

Authorized

Google Sheets access is active (OAuth token). — Disconnect

Pick Accounts

Search Logfood and add accounts to your demand plan

84.51 — Search

1 results

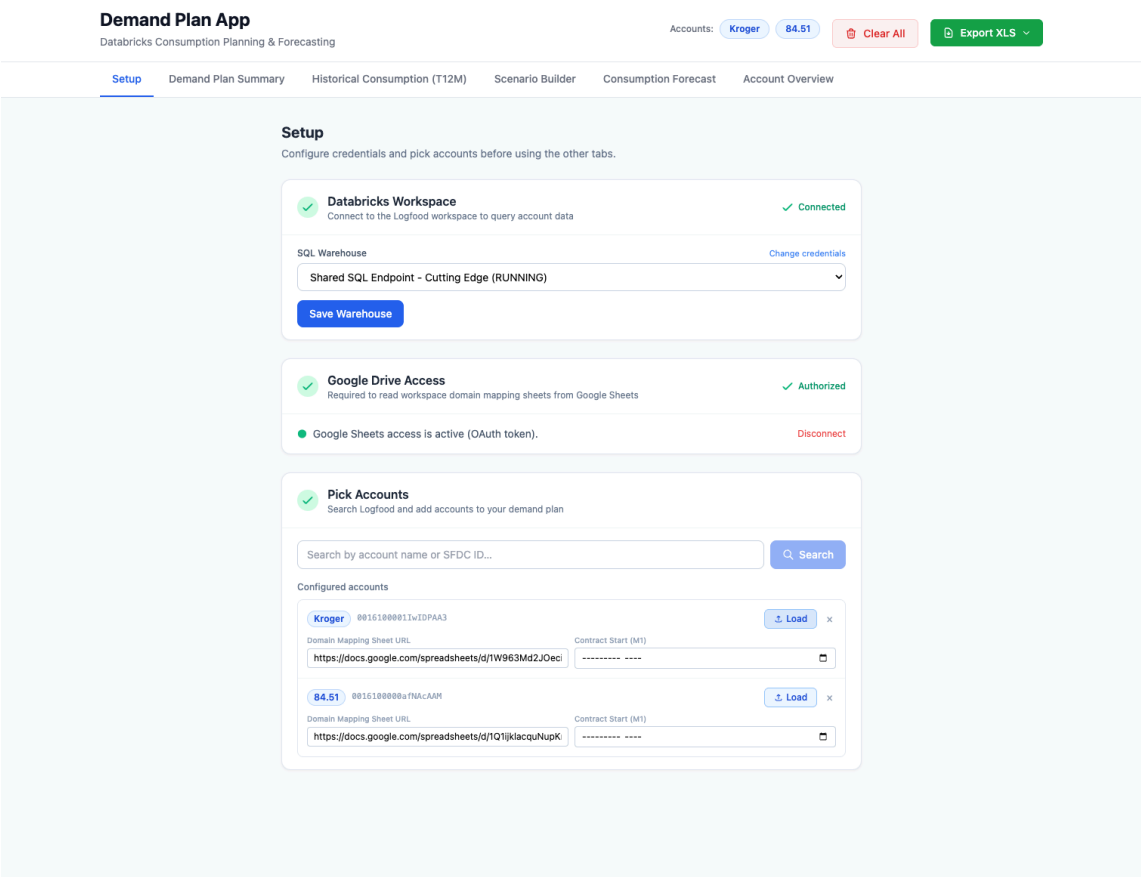
84.51
0016100000afNAcAAM

Configured accounts

Kroger 0016100001IwIDPAA3 — Load

Domain Mapping Sheet URL: https://docs.google.com/spreadsheets/...

Contract Start (M1): --------- ----



Loading Account Data

After adding accounts, click the **Load** button next to each account. This queries the logfood Databricks workspace for T12M historical consumption data, domain breakdowns, and SKU details.

Demand Plan App

Databricks Consumption Planning & Forecasting

Accounts: Kroger 84.51 | Clear All | Export XLS

Setup | Demand Plan Summary | Historical Consumption (T12M) | Scenario Builder | Consumption Forecast | Account Overview

Setup

Configure credentials and pick accounts before using the other tabs.

Databricks Workspace

Connect to the Logfood workspace to query account data

✓ Connected

SQL Warehouse — Change credentials

Shared SQL Endpoint - Cutting Edge (RUNNING)

Save Warehouse

Google Drive Access

Required to read workspace domain mapping sheets from Google Sheets

✓ Authorized

● Google Sheets access is active (OAuth token). — Disconnect

Pick Accounts

Search Logfood and add accounts to your demand plan

Search by account name or SFDC ID... | Search

Configured accounts

Kroger 0016100001IwIDPAA3 — Load ×

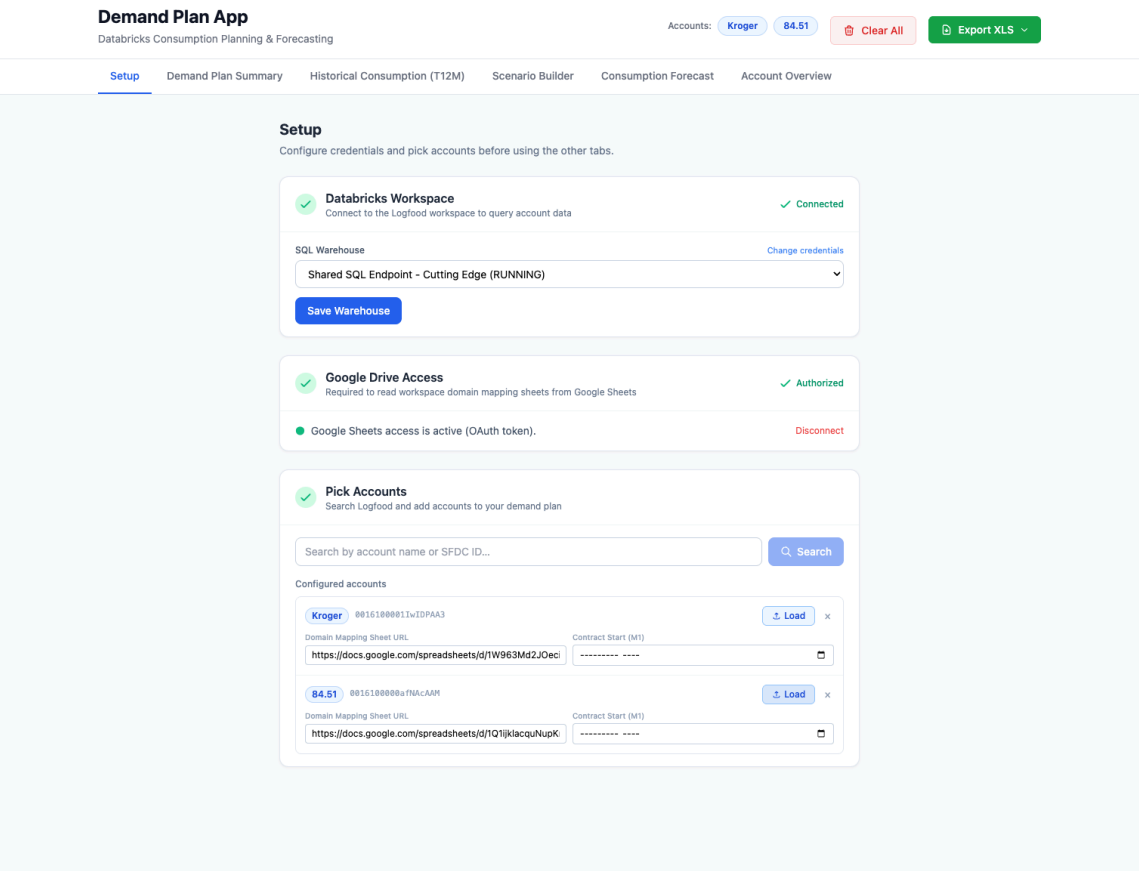
Domain Mapping Sheet URL: https://docs.google.com/spreadsheets/d/1W963Md2JOeci

Contract Start (M1): --------- ----

84.51 0016100000afNAcAAM — Load ×

Domain Mapping Sheet URL: https://docs.google.com/spreadsheets/d/1Q1ijklacquNupK

Contract Start (M1): --------- ----



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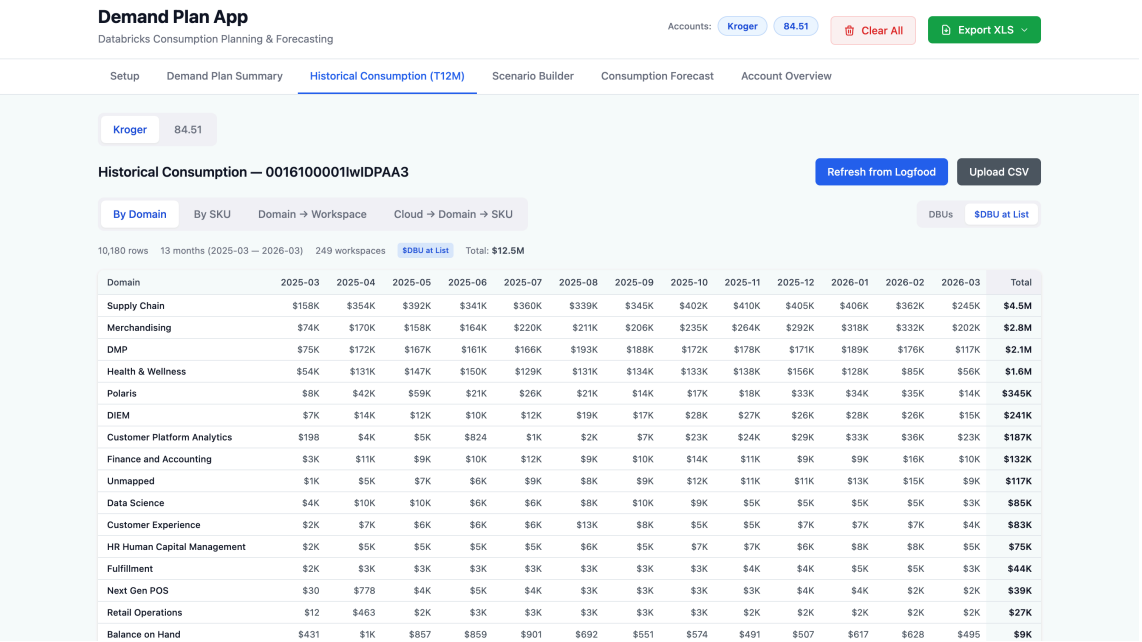
2. Historical Consumption (T12M)

This tab visualizes the trailing 12-month consumption for the selected accounts.

Features:

- **Monthly trend chart** — bar chart of $DBU spend per month
- **Domain breakdown table** — consumption by domain with T12M total, monthly average, and % share
- **SKU breakdown** — individual SKU-level spend within each domain
- **Account switcher** — toggle between loaded accounts using the top tab buttons



Domain	2025-03	2025-04	2025-05	2025-06	2025-07	2025-08	2025-09	2025-10	2025-11	2025-12	2026-01	2026-02	2026-03	Total
Supply Chain	$158K	$354K	$392K	$341K	$360K	$339K	$345K	$402K	$410K	$405K	$406K	$362K	$245K	**$4.5M**
Merchandising	$74K	$170K	$158K	$164K	$220K	$211K	$206K	$235K	$264K	$292K	$318K	$332K	$202K	**$2.8M**
DMP	$75K	$172K	$167K	$161K	$166K	$193K	$188K	$172K	$178K	$171K	$189K	$176K	$117K	**$2.1M**
Health & Wellness	$54K	$131K	$147K	$150K	$129K	$131K	$134K	$133K	$138K	$156K	$128K	$85K	$56K	**$1.6M**
Polaris	$8K	$42K	$59K	$21K	$26K	$21K	$14K	$17K	$18K	$33K	$34K	$35K	$14K	**$345K**
DIEM	$7K	$14K	$12K	$10K	$12K	$19K	$17K	$28K	$27K	$26K	$28K	$26K	$15K	**$241K**
Customer Platform Analytics	$198	$4K	$5K	$824	$1K	$2K	$7K	$23K	$24K	$29K	$33K	$36K	$23K	**$187K**
Finance and Accounting	$3K	$11K	$9K	$10K	$12K	$9K	$10K	$14K	$11K	$9K	$9K	$16K	$10K	**$132K**
Unmapped	$1K	$5K	$7K	$6K	$9K	$8K	$9K	$12K	$11K	$11K	$13K	$15K	$9K	**$117K**
Data Science	$4K	$10K	$10K	$6K	$6K	$8K	$10K	$9K	$5K	$5K	$5K	$5K	$3K	**$85K**
Customer Experience	$2K	$7K	$6K	$6K	$6K	$13K	$8K	$5K	$5K	$7K	$7K	$7K	$4K	**$83K**
HR Human Capital Management	$2K	$5K	$5K	$5K	$5K	$6K	$5K	$7K	$7K	$6K	$8K	$8K	$5K	**$75K**
Fulfillment	$2K	$3K	$3K	$3K	$3K	$3K	$3K	$3K	$4K	$4K	$5K	$5K	$3K	**$44K**
Next Gen POS	$30	$778	$4K	$5K	$4K	$3K	$3K	$3K	$3K	$4K	$4K	$2K	$2K	**$39K**
Retail Operations	$12	$463	$2K	$3K	$3K	$3K	$3K	$3K	$2K	$2K	$2K	$2K	$2K	**$27K**
Balance on Hand	$431	$1K	$857	$859	$901	$692	$551	$574	$491	$507	$617	$628	$495	**$9K**

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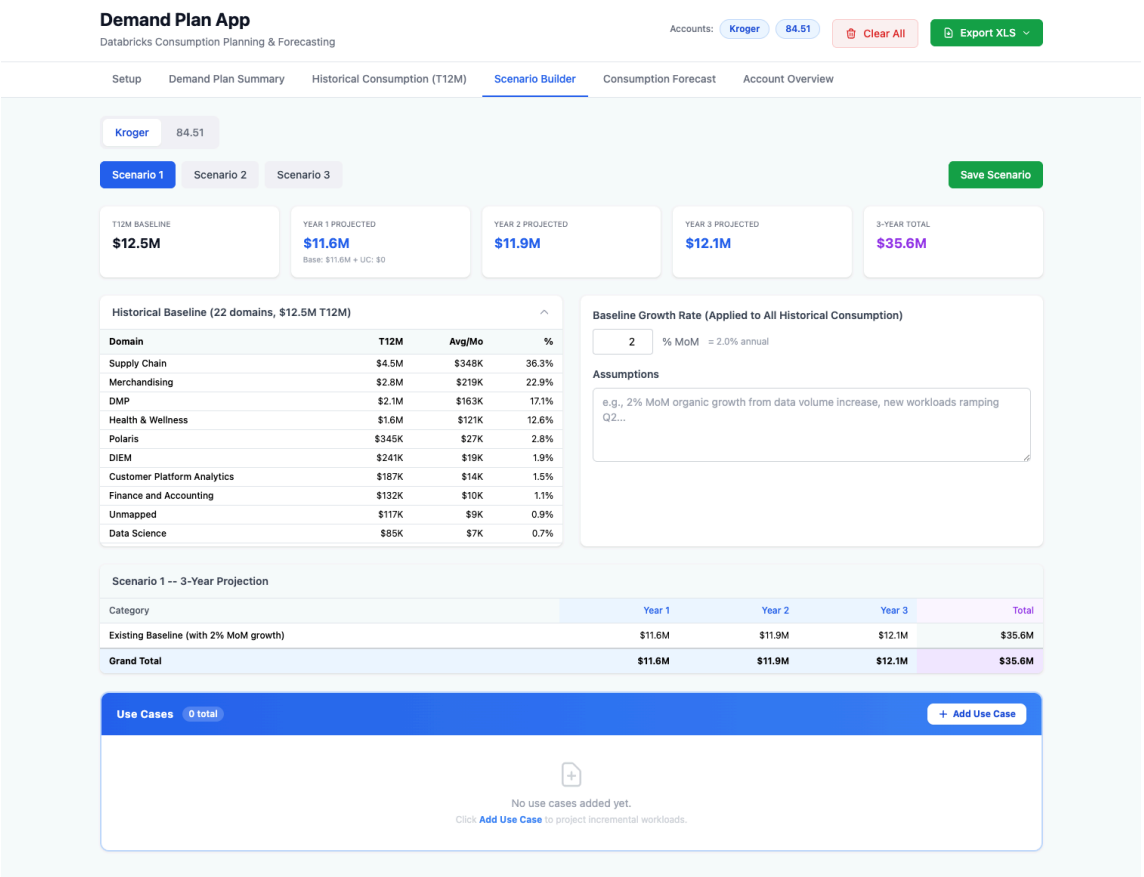
3. Scenario Builder

The Scenario Builder is the core planning tool. It lets you define 3 independent scenarios (S1, S2, S3) per account.



Structure

Each scenario has:

- **Baseline** — the T12M historical consumption with an optional MoM growth rate (default 2%)
- **Use Cases** — incremental workloads to add on top of the baseline
- **Assumptions** — free-text field for notes

Scenario Summary Panel

At the top of each scenario view, 4 key metrics are shown:

- T12M Baseline
- Year 1 Projected (Base + UC)
- Year 2 Projected
- Year 3 Projected
- 3-Year Total

The **3-Year Projection table** below shows the breakdown by use case.

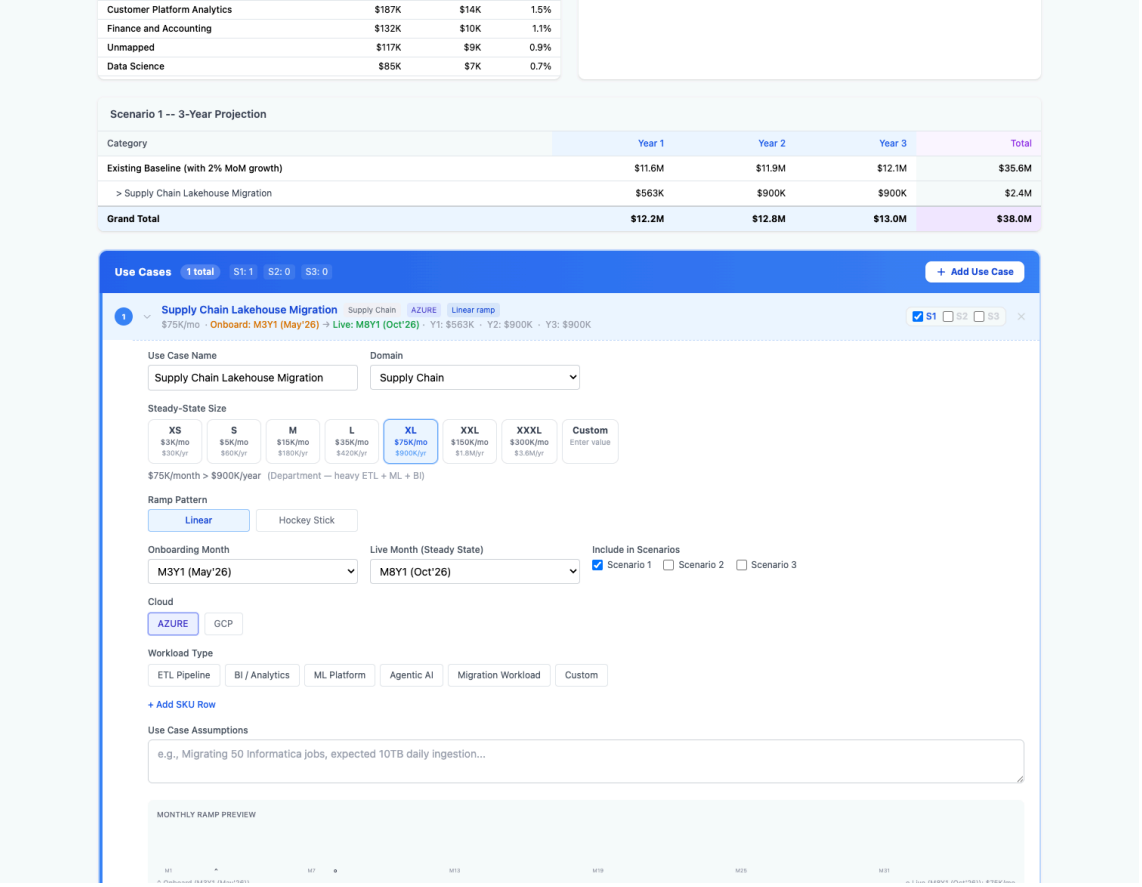
Adding Use Cases

1. Click **Add Use Case** — a new row appears in the use case list with default S1/S2/S3 assignment based on the current scenario view

2. Click the use case row to expand its edit form

3. Configure:
    - **Use Case Name** — e.g., "Supply Chain Lakehouse Migration"
    - **Domain** — select from existing historical domains
    - **Steady-State Size** — XS ($3K/mo) through XXXL ($300K/mo), or Custom
    - **Ramp Pattern** — Linear or Hockey Stick
    - **Onboarding Month** — when the use case starts ramping
    - **Live Month** — when full steady-state spend is reached
    - **Include in Scenarios** — checkboxes to assign the UC to S1, S2, and/or S3
    - **Cloud** — AWS or AZURE
    - **Workload Type** — ETL Pipeline, BI/Analytics, ML Platform, Agentic AI, Migration Workload, Custom
4. Click **Save Scenario** when done

The **Monthly Ramp Preview** at the bottom shows a visual ASCII chart of the ramp curve.



Category	Year 1	Year 2	Year 3	Total
Existing Baseline (with 2% MoM growth)	$11.6M	$11.9M	$12.1M	$35.6M
> Supply Chain Lakehouse Migration	$563K	$900K	$900K	$2.4M
> Merchandising AI & Analytics Platform	$263K	$420K	$420K	$1.1M
Grand Total	**$12.5M**	**$13.2M**	**$13.4M**	**$39.1M**

Use Cases 2 total S1: 2 S2: 0 S3: 0 + Add Use Case

1 **Supply Chain Lakehouse Migration** Supply Chain AZURE Linear ramp
$75K/mo · Onboard: M3Y1 (May'26) → Live: M8Y1 (Oct'26) · Y1: $563K · Y2: $900K · Y3: $900K
☑ S1 ☐ S2 ☐ S3

2 **Merchandising AI & Analytics Platform** Merchandising AZURE Linear ramp
$35K/mo · Onboard: M3Y1 (May'26) → Live: M8Y1 (Oct'26) · Y1: $263K · Y2: $420K · Y3: $420K
☑ S1 ☐ S2 ☐ S3

Use Case Name: Merchandising AI & Analytics Platform
Domain: Merchandising

Steady-State Size

XS	S	M	L	XL	XXL	XXXL	Custom
$3K/mo $30K/yr	$5K/mo $60K/yr	$15K/mo $180K/yr	$35K/mo $420K/yr	$75K/mo $900K/yr	$150K/mo $1.8M/yr	$300K/mo $3.6M/yr	Enter value

$35K/month > $420K/year (Multi-pipeline ETL + ML)

Ramp Pattern: Linear | Hockey Stick

Onboarding Month: M3Y1 (May'26)
Live Month (Steady State): M8Y1 (Oct'26)
Include in Scenarios: ☑ Scenario 1 ☐ Scenario 2 ☐ Scenario 3

Cloud: AZURE | GCP

Workload Type: ETL Pipeline | BI / Analytics | ML Platform | Agentic AI | Migration Workload | Custom

+ Add SKU Row

Use Case Assumptions
e.g., Migrating 50 Informatica jobs, expected 10TB daily ingestion...

MONTHLY RAMP PREVIEW
M1 M7 M13 M19 M25 M31
^ Onboard (M3Y1 (May'26)) o Live (M8Y1 (Oct'26)): $35K/mo

3 **DMP Personalization Engine** DMP AZURE Linear ramp
$15K/mo · Onboard: M3Y1 (May'26) → Live: M8Y1 (Oct'26) · Y1: $113K · Y2: $180K · Y3: $180K
☐ S1 ☑ S2 ☐ S3

Use Case Name: DMP Personalization Engine
Domain: DMP

Steady-State Size

XS	S	M	L	XL	XXL	XXXL	Custom
$3K/mo $30K/yr	$5K/mo $60K/yr	$15K/mo $180K/yr	$35K/mo $420K/yr	$75K/mo $900K/yr	$150K/mo $1.8M/yr	$300K/mo $3.6M/yr	Enter value

$15K/month > $180K/year (Team workload — ETL + BI)

Ramp Pattern: Linear | Hockey Stick

Onboarding Month: M3Y1 (May'26)
Live Month (Steady State): M8Y1 (Oct'26)
Include in Scenarios: ☐ Scenario 1 ☑ Scenario 2 ☐ Scenario 3

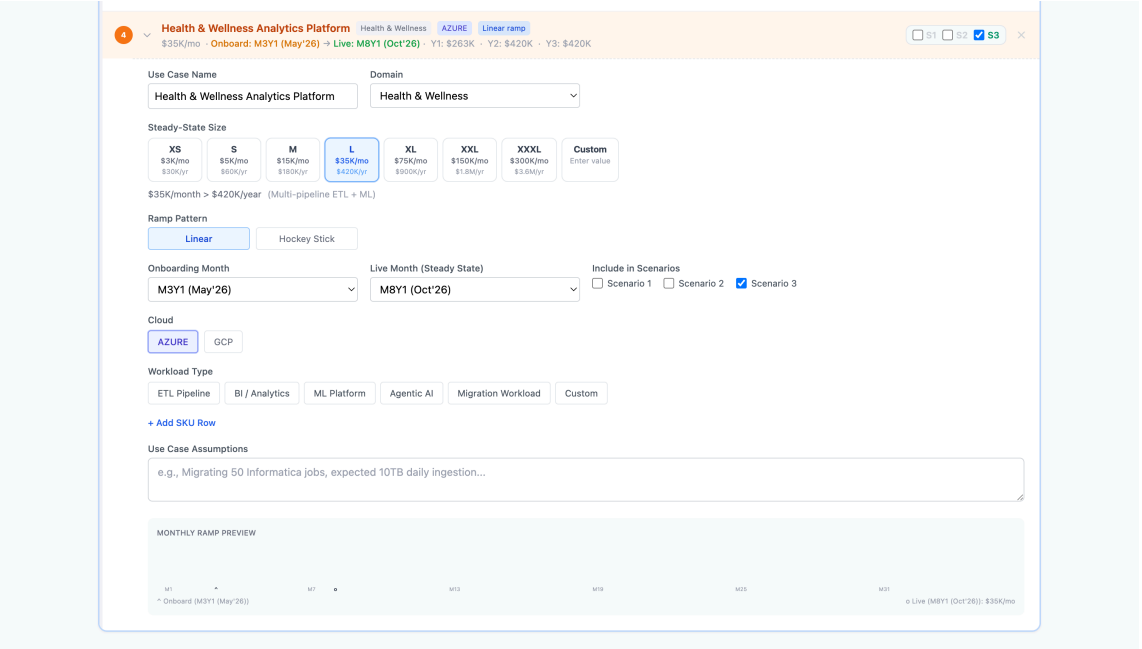
Cloud: AZURE | GCP

Workload Type: ETL Pipeline | BI / Analytics | ML Platform | Agentic AI | Migration Workload | Custom

+ Add SKU Row

Use Case Assumptions
e.g., Migrating 50 Informatica jobs, expected 10TB daily ingestion...

MONTHLY RAMP PREVIEW
M1 M7 M13 M19 M25 M31
^ Onboard (M3Y1 (May'26)) o Live (M8Y1 (Oct'26)): $15K/mo



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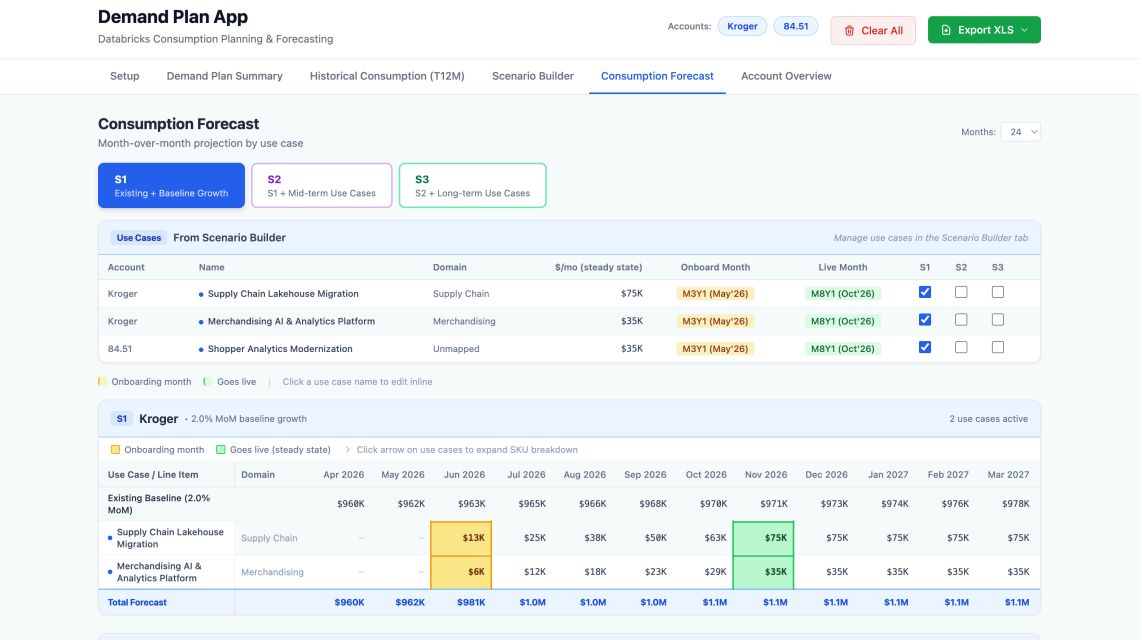
4. Consumption Forecast

The Consumption Forecast tab shows a month-by-month projection table built from the last **saved** scenario state.

> ***Note:*** *You must click* ***Save Scenario*** *in the Scenario Builder before the forecast reflects your latest changes.*

Features:

- Month columns from the contract start date through Year 3
- Rows per use case with monthly dollar amounts
- Baseline row showing organic growth
- Grand total row



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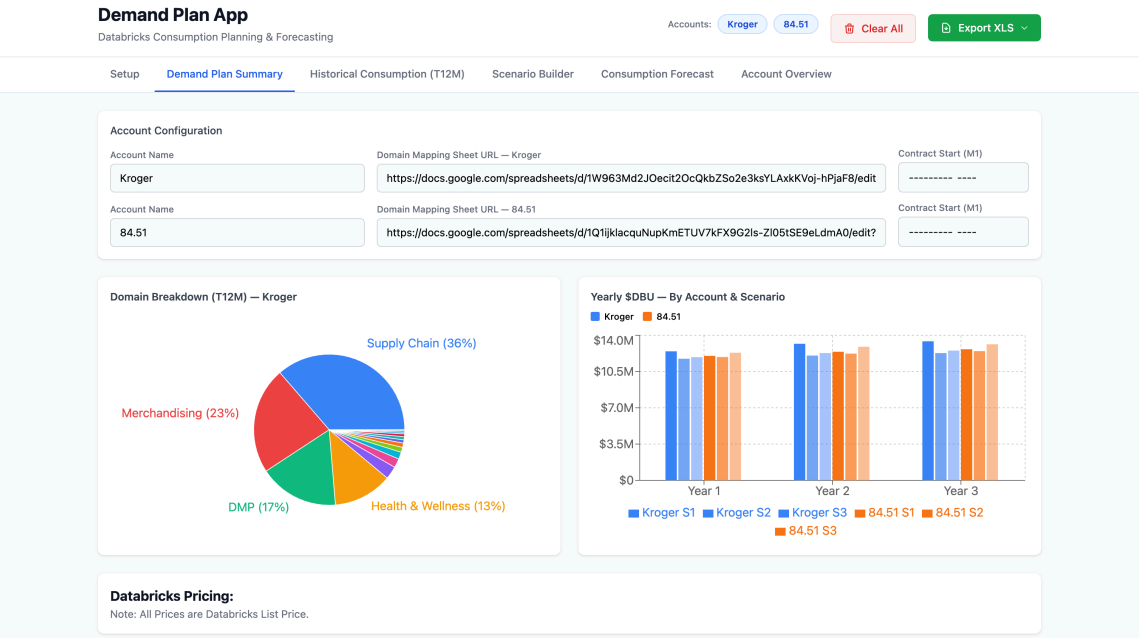
5. Demand Plan Summary

The Demand Plan Summary aggregates projections across **all loaded accounts** and all 3 scenarios.

Features:

- Combined 3-year table (all accounts × S1/S2/S3)
- Per-account breakdown with Year 1/2/3 and Total columns
- Scenario toggle (S1 / S2 / S3 buttons)
- Use case contributions shown separately from baseline

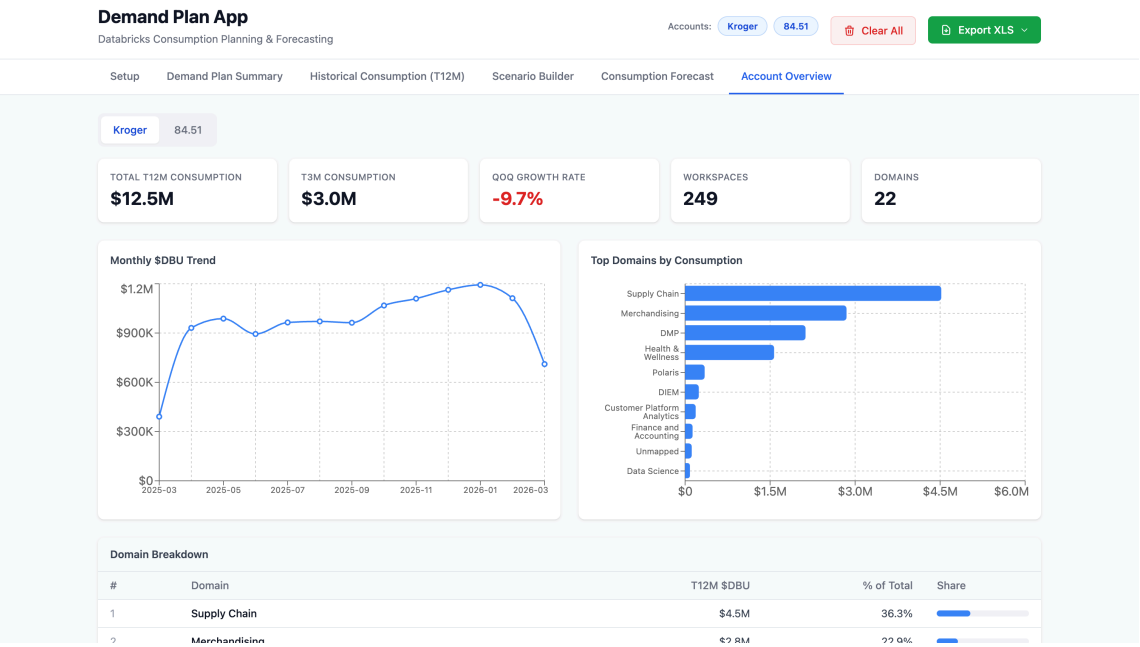


6. Account Overview

The Account Overview tab provides a high-level view of each account's historical consumption profile.

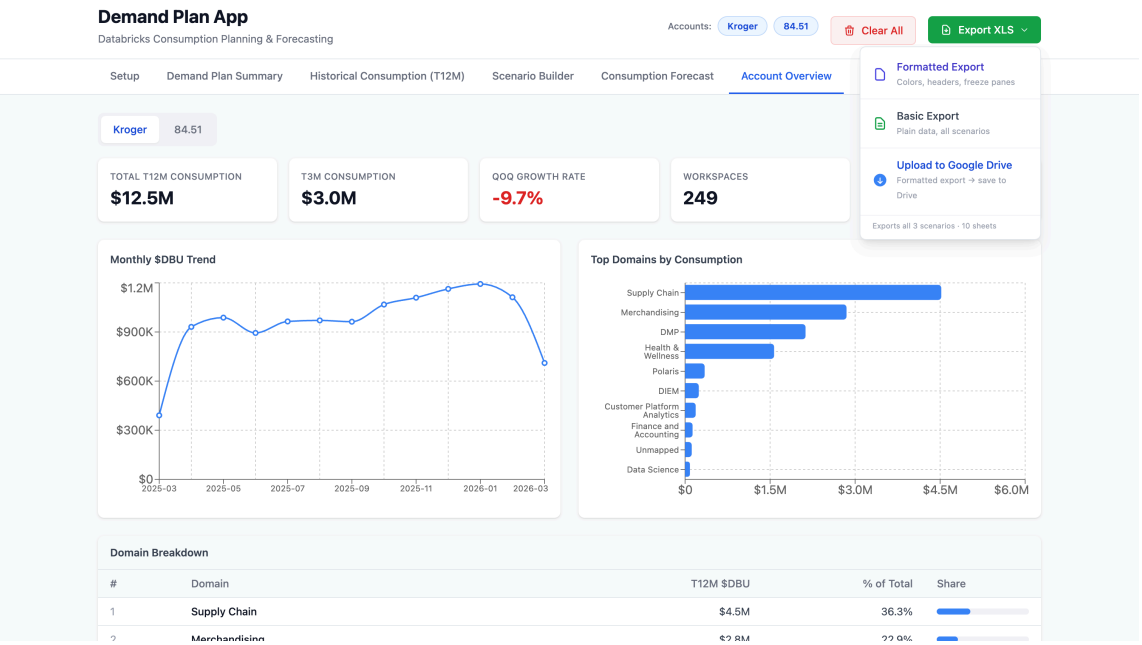
Features:

- KPI cards: Total T12M, T3M, QoQ Growth Rate, Workspaces, Domains
- **Monthly $DBU Trend** — 12-month bar chart
- **Top Domains by Consumption** — horizontal bar chart
- **Domain Breakdown** table with T12M $DBU and % share



7. Export

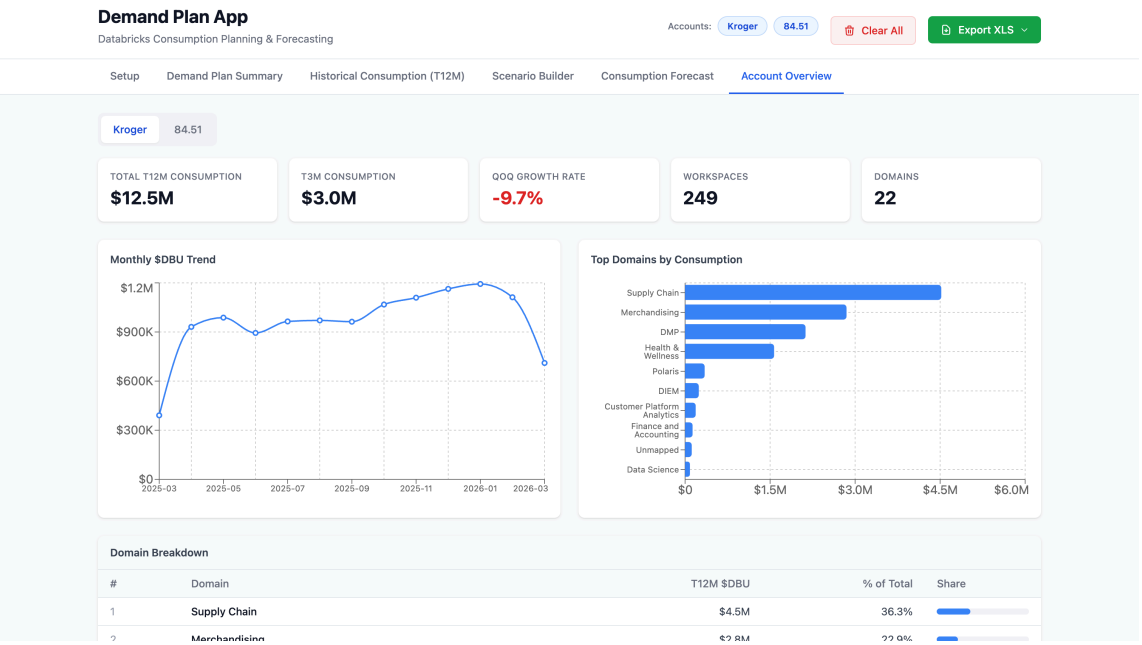
Click the **Export XLS** button (top-right) to open the export menu with 3 options:



Formatted Export

Generates a multi-sheet Excel workbook with:

- Colors, headers, freeze panes
- **Demand Plan Summary** sheet
- **S1 / S2 / S3 — Projection** sheets (real calendar month headers, all accounts)
- Per-account **Hist-Domain**, **Hist-SKU**, **Cloud-Domain-SKU** sheets
- Per-account **Use Cases** sheet with SKU sub-rows
- Per-account **Baseline** and **SKU Detail** sheets



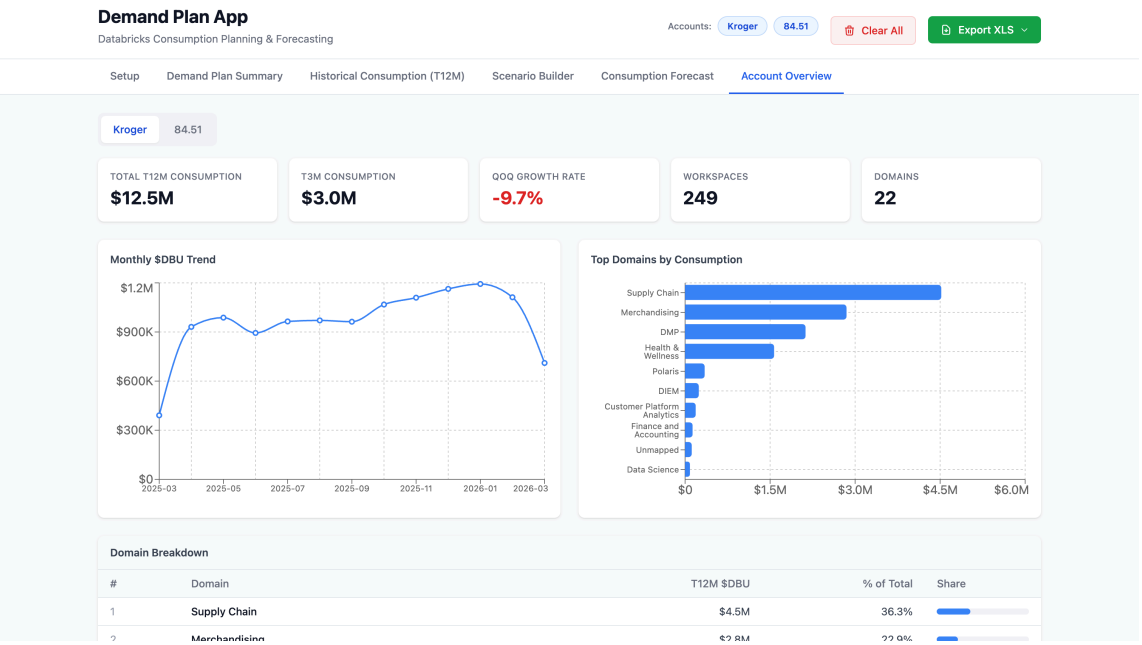
Basic Export

Plain data export, all scenarios, no formatting.

Upload to Google Drive

Builds the formatted Excel workbook and uploads it to Google Drive via the backend. On success, a toast notification appears with:

- Upload confirmation
- File name
- Google Drive URL (copyable monospace box)
- "Open in Drive →" link



Use Case Testing — Kroger & 84.51

The following use cases were added across all 3 scenarios for both accounts.

Kroger (`0016100001IwIDPAA3`)

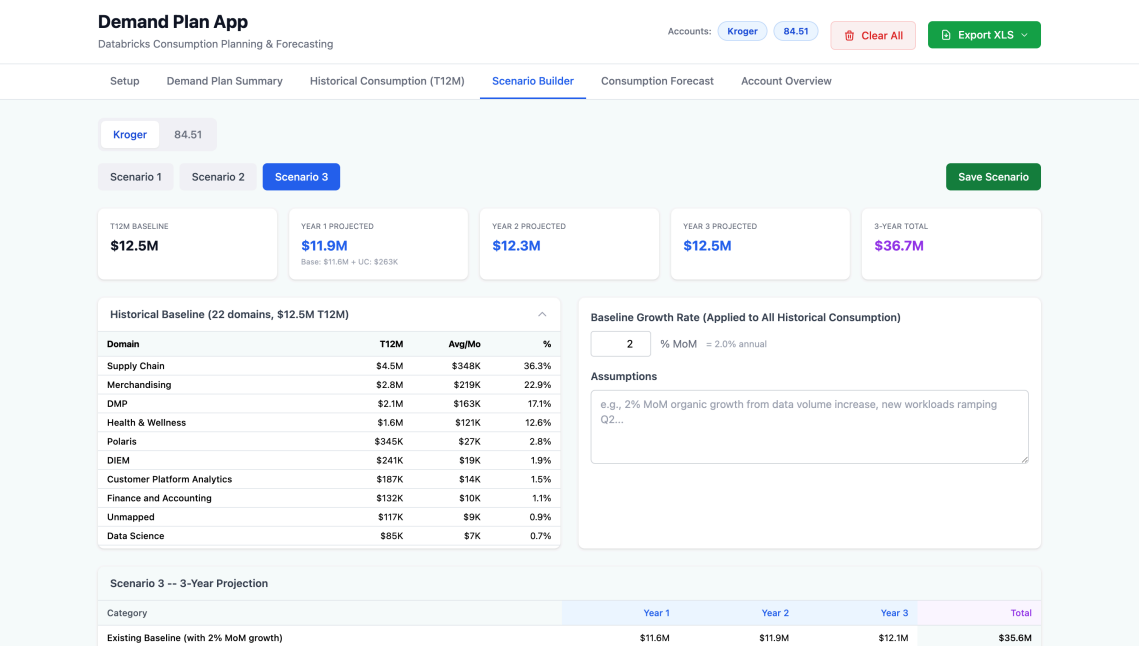
Domain Mapping: Kroger Google Sheets

#	Use Case	Domain	Size	Scenario	Ramp	Y1	Y2	
1	Supply Chain Lakehouse Migration	Supply Chain	XL $75K/mo	**S1**	Linear	$563K	$900K	$9
2	Merchandising AI & Analytics Platform	Merchandising	L $35K/mo	**S1**	Linear	$263K	$420K	$4

3	DMP Personalization Engine	DMP	M $15K/mo	**S2**	Linear	$113K	$180K	$1
4	Health & Wellness Analytics Platform	Health & Wellness	L $35K/mo	**S3**	Linear	$263K	$420K	$4

Scenario Summary (Kroger):

Scenario	Y1	Y2	Y3	Total
S1 (Baseline + UC1 + UC2)	~$12.4M	~$13.2M	~$13.5M	~$39.1M
S2 (Baseline + UC3)	~$11.7M	~$12.0M	~$12.3M	~$36.1M
S3 (Baseline + UC4)	~$11.9M	~$12.3M	~$12.5M	~$36.7M



84.51 (`0016100000afNAcAAM`)

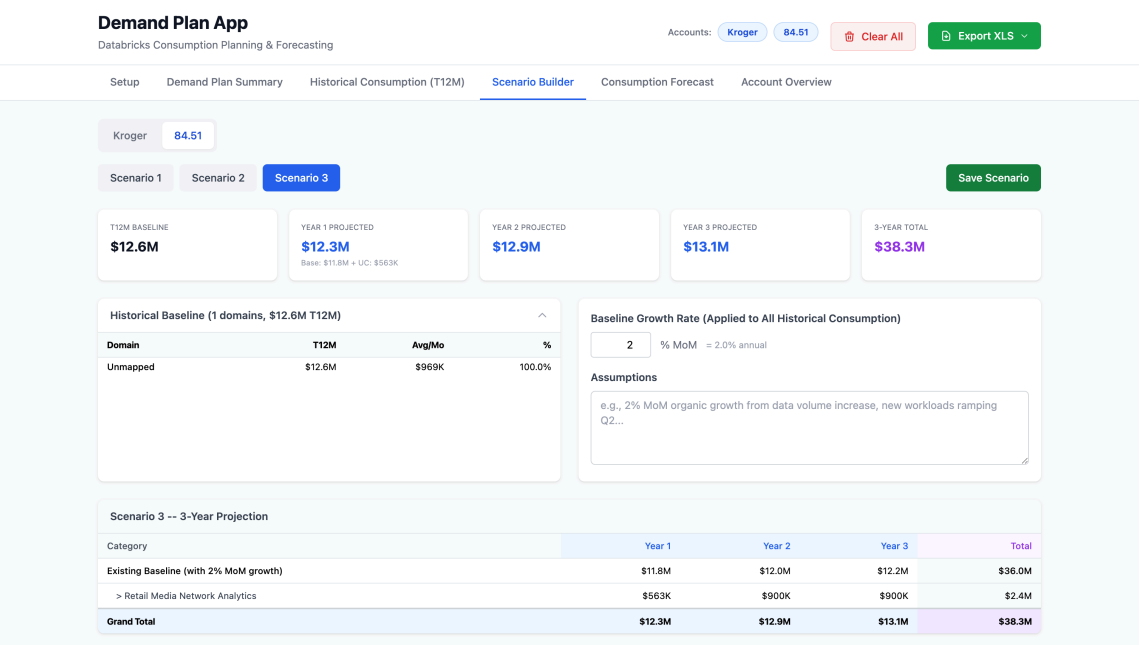
Domain Mapping: 84.51 Google Sheets

#	Use Case	Domain	Size	Scenario	Ramp	Y1	Y2	Y3
1	Shopper Analytics Modernization	Unmapped	L $35K/mo	**S1**	Linear	$263K	$420K	$420
2	Real-Time Personalization	Unmapped	M $15K/mo	**S2**	Linear	$113K	$180K	$180

	Engine							
3	Supplier Intelligence Platform	Unmapped	S $5K/mo	**S2**	Linear	$38K	$60K	$60K
4	Retail Media Network Analytics	Unmapped	XL $75K/mo	**S3**	Linear	$563K	$900K	$900K

Scenario Summary (84.51):

Scenario	Y1	Y2	Y3	Total
S1 (Baseline + UC1)	~$12.1M	~$12.4M	~$12.7M	~$37.1M
S2 (Baseline + UC2 + UC3)	~$11.9M	~$12.2M	~$12.5M	~$36.6M
S3 (Baseline + UC4)	~$12.3M	~$12.9M	~$13.1M	~$38.4M



Export Results

Formatted Excel Export

The formatted export was downloaded successfully as:

```
Demand_Plan_Kroger_84.51_AllScenarios_2026-03-19.xlsx
```

Sheets included:

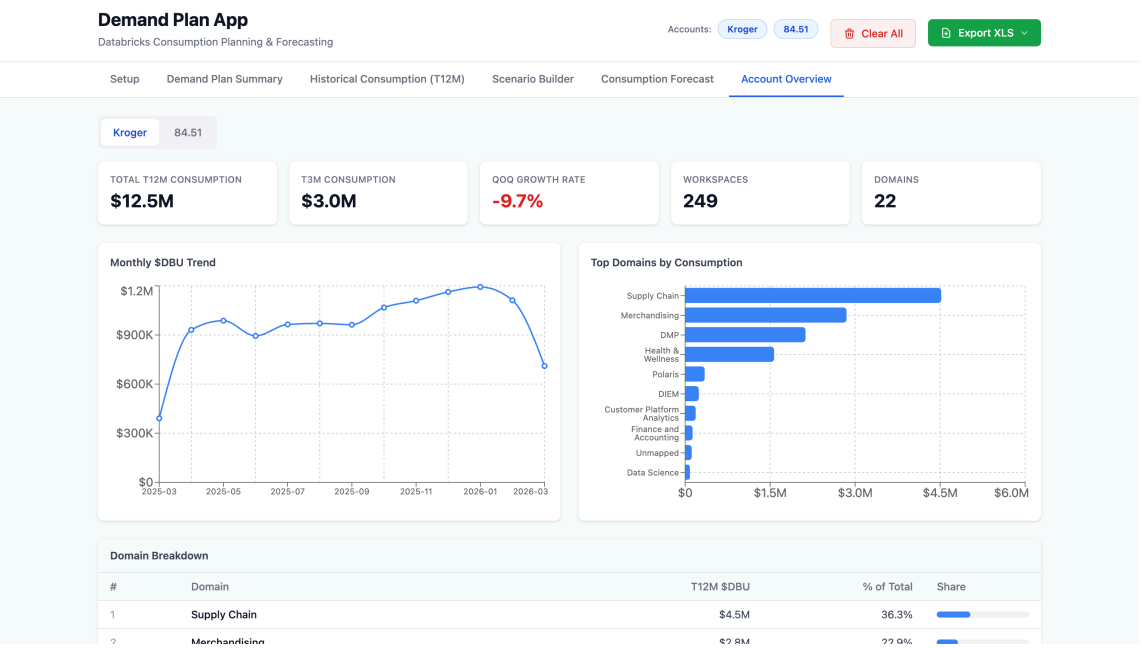
Sheet	Description

Demand Plan Summary	Combined across all accounts
S1 - Projection	Scenario 1: all accounts, real calendar month headers
S2 - Projection	Scenario 2: same structure
S3 - Projection	Scenario 3: same structure
Kroger Hist-Domain	Historical by domain
Kroger Hist-SKU	Historical by SKU
Kroger Cloud-Domain-SKU	Cloud/domain/SKU breakdown
Kroger Use Cases	Use case details with SKU sub-rows
Kroger Baseline	Baseline consumption
Kroger SKU Detail	SKU-level detail
84.51 Hist-Domain	Historical by domain
84.51 Hist-SKU	Historical by SKU
84.51 Cloud-Domain-SKU	Cloud/domain/SKU breakdown
84.51 Use Cases	Use case details
84.51 Baseline	Baseline consumption
84.51 SKU Detail	SKU-level detail

Google Drive Upload

Excel uploaded to Google Drive:

- **File:** `Demand_Plan_Kroger_84.51_AllScenarios_2026-03-19.xlsx`
- **Drive URL:** https://drive.google.com/file/d/1ThZrK5vRU04nI0vT6kt0amZTudju3Hqu/view



Key Features Summary

Feature	Description
Default Setup Tab	App opens on Setup, not Summary
Account Search	Live search with × close button and auto-close on add
Per-Account Load	Independent Load button per account
Domain Mapping URL	Google Sheets URL stored per account for SKU → domain resolution
Multi-Scenario Planning	3 independent scenarios (S1/S2/S3) per account
Use Case Assignment	Each use case can be included in any combination of S1/S2/S3
Real Calendar Headers	Export uses `May_2026`, `Jun_2026` format (not M1/M2)
Multi-Account Projections	All accounts appear as sections in shared S1/S2/S3 sheets
Formatted Excel Export	Colors, freeze panes, SKU sub-rows, per-account sheets
Google Drive Upload	Backend multipart upload with Drive URL in success toast
Clear All	Resets all accounts and cached data to blank state